Profiling GRPO vs. turn-PPO on ASearcher (Qwen3-8B-Base): 1×H100 node, 1×B200 node, and the 2-node verdict

CA-benchmark rung-3 / 8B thread

2026-08-04

Setup

Both algorithms train Qwen3-8B-Base on unfiltered ASearcher Base-35k (wiki-18 / e5 top-5 retrieval, 32-turn horizon, 16k prompt / 1024 response) under the *fast* config: partial rollout (cycle 8 / max-age 4), dynamic token batching, chunked prefill; batch 256×5, PPO mini-batch 512, LR 10^{-6}. Data sources:

- **1×H100**: the two *finished 75-step arms* (not short profiling runs) — per-phase `timing_s/*` logged every step. turn-PPO ran entirely at DYNBSZ 24576 (critic FSDP-offloaded); GRPO ran steps 1–25 at 24576, then 26–75 at 20480 after an OOM.
- **1×B200**: fresh 8-step runs of each algorithm on one 8×B200 node, DYNBSZ 24576, critic *on-GPU* (192 GB), torch flat retrieval (no sm_100 faiss). Position-matched to H100 steps 1–N, which ran at the same DYNBSZ.
- **2×H100**: no run is possible; measured evidence in §4.

Comparability: model, data, sampler settings, rollout config and batch geometry identical across hardware; the two deliberate deltas are critic placement (offload vs. on-GPU — the B200 memory advantage under test) and the retrieval server implementation (identical HTTP contract; retrieval is off the critical path, <1% of step time). Because sampled trajectories differ slightly across hardware, tables report both wall-clock and *per-token* ratios; per-token is the transferable number.

Instrumentation limits. verl does not split forward/backward/optimizer inside `update_*` (reported as one unit; actor MFU ≈0.31 on H100 both arms, critic 0.32), and FSDP↔vLLM weight resharding is contained in `gen`. Data loading and preprocessing are in "other" (<0.5% throughout).

1 1×H100, full 75-step runs

GRPO — 65.9 h in-step, 3,138M tokens, 13.2k tok/s phase	time	share
rollout generation	30.8 h	46.7%
actor update	22.6 h	34.3%
old logprob	5.2 h	7.9%
ref policy (KL)	5.2 h	7.9%
val + ckpt + reward/adv/other	4.0 h	6.1%
TOTAL (in-step)	65.9 h	100%

turn-PPO — 72.3 h in-step, 2,347M tokens, 9.0k tok/s phase	time	share
rollout generation	30.0 h	41.4%
actor update	17.0 h	23.5%
critic update	16.2 h	22.3%
critic fwd (values)	3.6 h	5.0%
old logprob	3.9 h	5.3%
val + ckpt + reward/adv/other	3.3 h	4.6%
TOTAL (in-step)	72.3 h	100%

Table 1: Full-run phase totals on one H100 node. Wall-clock including crash-recovery was 73.1 h (turn-PPO, single clean attempt) and 89.4 h (GRPO, one OOM + resume).

Bottlenecks (H100). GRPO: rollout generation, 46.7% of the run (actor update 34.3%, logprob+ref 15.8%). turn-PPO: the update side — actor+critic updates plus critic forward total 50.8%, overtaking gen (41.4%). The single biggest lever for both remains active-slot packing in gen (finished trajectories still generate every round).

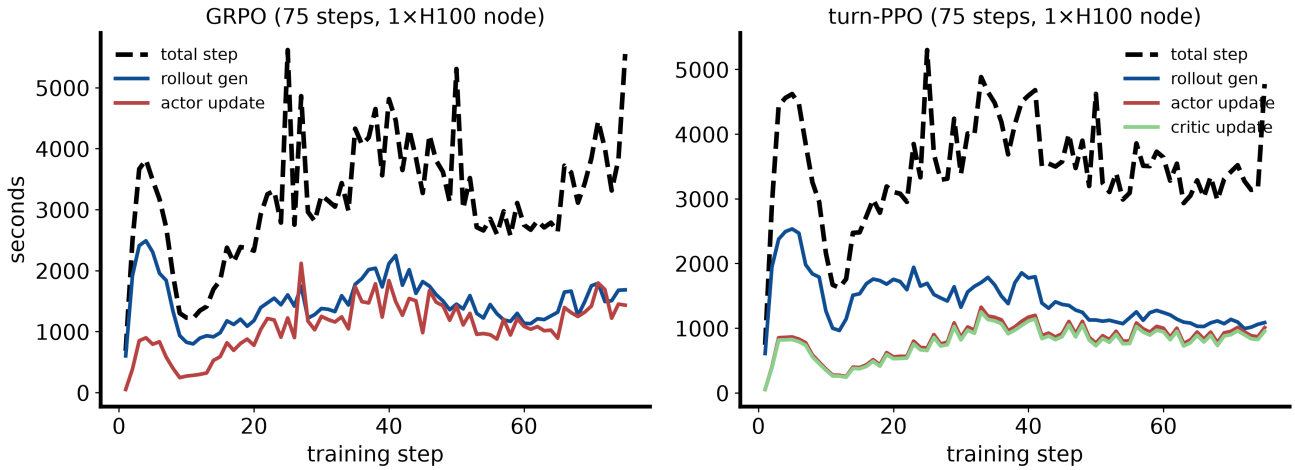

Figure 1: Per-step phase times across training. GRPO's step time grows 1.6× (2,447 s first-10 → 3,885 s last-10): its policy inflates to ∼17 turns and per-turn responses shrink, so update + logprob costs triple while gen stays flat. turn-PPO is flat end-to-end (3,395 s → 3,393 s): the critic steers toward ∼7.5-turn episodes, gen *falls* from 1,930 s to 1,067 s, paying for the critic's own ∼27% overhead. Spikes are val/checkpoint steps.

2 H100 vs. B200, position-matched steps

turn-PPO, matched early steps, mean						
phase	H100 s	share	B200 s	share	wall ratio	per-token
rollout gen	2057	56.3%	1787	64.1%	1.15×	1.08×
old logprob	149	4.1%	96	3.4%	1.56×	1.46×
critic fwd	141	3.9%	91	3.3%	1.55×	1.45×
critic update	631	17.3%	394	14.1%	1.60×	1.50×
actor update	661	18.1%	408	14.6%	1.62×	1.52×
TOTAL / step	3654	100%	2787	100%	1.31×	1.23×
tokens/step	24.8M		23.3M			

GRPO, matched early steps, mean						
phase	H100 s	share	B200 s	share	wall ratio	per-token
(B200 run in progress)						

Table 2: Position-matched phase means. H100 critic is FSDP-offloaded; B200 critic on-GPU — that placement difference is included deliberately as "what you would actually run" on each hardware.

Reading. Training-side phases (updates, logprob/values forwards) gain ∼1.5–1.6× on B200 — bf16 GEMM headroom (2.1× microbench) discounted by non-GEMM time at MFU ≈0.31. Generation gains only ∼1.1–1.2×: vLLM on sm_100 still runs the TRITON attention backend (vendored flash-attn segfaults in CUDA-graph capture), and decode is memory/latency-bound rather than GEMM-bound. Since gen is 41–47% of these workloads, end-to-end lands well below the GEMM ratio.

Full-run estimate. Dividing each H100 full-run phase total by its measured per-token ratio (val/checkpoint kept at 1×; identical token trajectory assumed): 75-step turn-PPO ≈ 56.6 h (72.3 h on H100, 1.28×); 75-step GRPO ≈ (pending). This matches the 1.53× end-to-end measured in the 2026-08-01 four-cycle pilot within its noise. An 8-step pilot cannot capture policy-drift effects on gen; treat these as ±10% estimates.

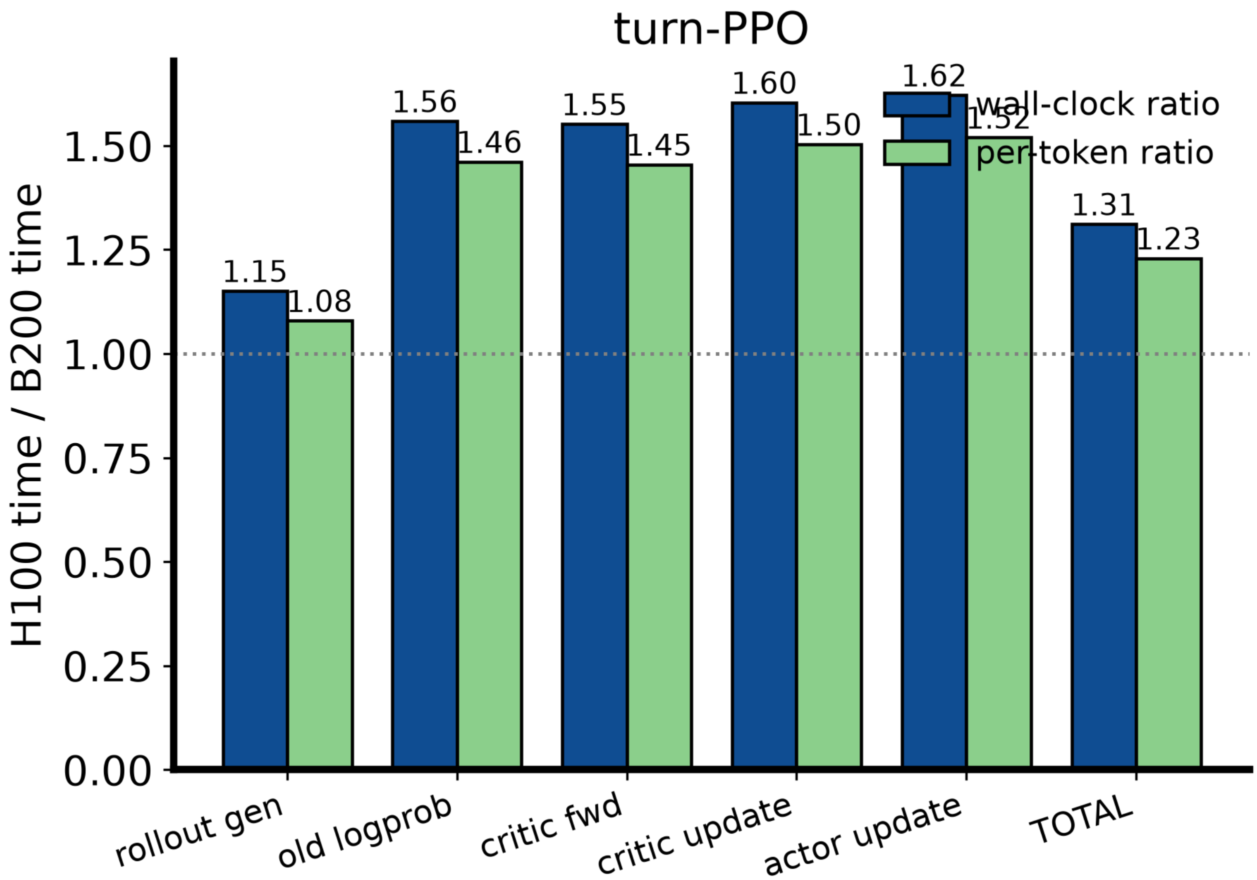

Figure 2: H100→B200 speedup by phase (left bars wall-clock, right bars per-token).

3 Cost perspective

At equal node counts B200 shortens a 75-step arm from ∼3 days to ∼2. Caveats that are operational, not performance: cross-region HDFS staging (25–360 MB/s, ∼25 min/pod), no faiss (torch retrieval server), no egress (wandb/HF offline), and the LD_PRELOAD libcuda fix. All are scripted in `p8b_b200_prof_worker.sh`.

4 2×H100: measured verdict — not runnable on current resources

Two independent blockers, both verified:

1. **Workspace workers** (our queue): no `/dev/infiniband`; NCCL falls back to socket transport and its ephemeral data ports are firewalled — the first cross-node allreduce fails with `ncclRemoteError` (probe `nccl_probe.sh`, 2026-08-03). Rendezvous and Ray cluster formation work; only collectives fail.
2. **Arnold training jobs**: per the platform's resource-queue manual, RDMA requires a *minipod*-tagged queue; the `mlsys_inference` group has none (checked 2026-08-04). Best case is socket NCCL at single-digit GB/s vs. 400 GB/s NVLink intra-node — with updates at 34–51% of step time, 2 nodes would be slower than 1.

Unblock: a minipod-bound H100 queue for `mlsys_inference` (the platform's own term for RDMA-provisioned training queues). A ready-to-submit 2-node probe job (`scripts/asearcher/mlx_nccl_probe_job.yaml`) verifies transport the day such a queue exists. No honest scaling-efficiency number can be produced before then.

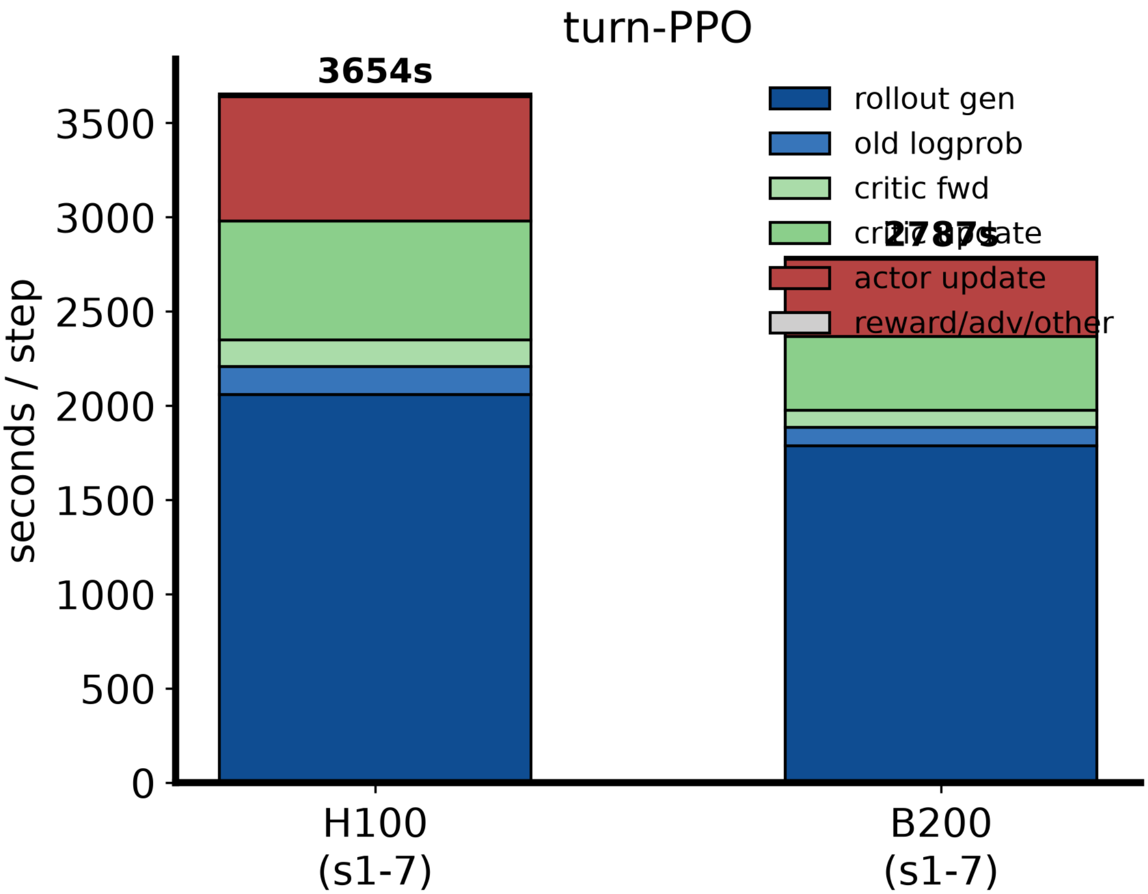

Figure 3: Step composition, matched steps.

Reproduction

`python mine_timings.py --out assets/timings.json grpo_h100=... turn_ppo_h100=... b200=...` then `python make_figures.py` and `tectonic hw_algo_profiling.tex`. Raw per-step data: `assets/timings.json`; computed numbers: `assets/computed_tables.json`.